

Threshold-Aware Similarity Caching for Named Data Networks: Bounding Retry Delay under Congestion using ns-3

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Abstract

Background

Content caching is one of the main techniques used to deliver content quickly and to reduce the traffic carried by a network. Content Delivery Networks, mobile edge caches and the in-network caches of Named Data Networking (NDN) all follow the same rule: a cache serves a request only when it holds the exact object that was asked for [1]. Similarity caching relaxes this rule. On a miss, the cache may return the most similar object it holds, which is useful for applications such as image retrieval, recommendation systems and semantic caching of large language model queries [2], [3]. The user accepts the returned object only if it is similar enough, and otherwise sends the request again.

State-of-the-Art

Nakamura and Kamiyama recently analysed similarity caching on a general cache network and introduced the similar-content delivery delay, a user-level metric that includes the time spent on repeated requests [4]. In their model each cache answers with a fixed probability α , and the network is assumed to have infinite bandwidth, no queuing and no packet loss. The results are obtained only through numerical evaluation on a tandem topology of five caches. Other studies consider a single similarity cache or optimise content placement in networks of similarity caches [2], [5], but none of them evaluates the request and retry behaviour at the packet level.

Issues and Challenges

Several challenges exist in making similarity caching work in a real multi-hop network:

- **Correlated Retries:** The analysis treats every retry as an independent draw of the cache contents. In practice the contents do not change within the retry interval, so a user who rejects an answer keeps receiving the same one. Re-evaluating the model on the same topology with persistent cache contents, we found that the mean delay is underestimated by about two times at $\alpha = 0.5$ and by more than sixty times at $\alpha = 0.9$.

- **Blind Caches:** A cache does not know the similarity that the user will accept, so it can return an object that is certain to be rejected.
- **NDN Name Matching:** NDN delivers a Data packet only when its name matches the Interest, so returning a different but similar object needs changes to the packet format and the forwarder.
- **Congestion and Loss:** The benefit of answering from a nearby cache depends on the queuing delay and loss on the upstream path, which the existing model does not capture.

Objectives

This project aims to design and implement threshold-aware similarity caching for NDN and to evaluate it in ns-3 using the ndnSIM module [6]. The specific objectives are:

1. Implement a similarity-based Content Store along with consumer and producer applications in ndnSIM, and reproduce the results of the base paper on the same tandem topology.
2. Measure the retry delay of blind similarity caching when the cache contents persist across requests, and compare it with the analytical model.
3. Design an Interest format that carries the minimum similarity accepted by the user and a bound on the number of retries, so that a cache answers only when its best match is acceptable.
4. Design a forwarding strategy that chooses between answering locally and forwarding based on the measured upstream delay, and evaluate it under link congestion and under burst packet loss.
5. Compare the proposed scheme with the base scheme in terms of mean and 95th percentile delivery delay, delivered similarity, upstream traffic and fairness across clients.

The project will show whether simple changes to the request format and to the decision rule at each cache can make similarity caching dependable in multi-hop networks, where a wrong answer from a nearby cache costs the user a complete retry.

Methodology Overview

The proposed system architecture consists of:

- ns-3 with the ndnSIM module as the simulation platform
- Similarity-based Content Store replacing exact name lookup at each cache node
- Consumer applications that check the similarity and retry, and a producer acting as the origin server
- Custom forwarding strategy implementing the answer-or-forward decision
- Point-to-point links with configurable bandwidth, delay, queue size and burst loss
- Python and MATLAB scripts comparing the simulation results with the analytical model

Team Details

- **Somyak Priyadarshi Mohanta** (241CS257) - Implementation lead, Content Store and forwarding strategy, system architecture
- **Pranav Shaji** (241CSXXX) - Consumer and producer applications, simulation scenarios
- **Chris Tony** (241CSXXX) - Analytical model, performance analysis, documentation

References

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